Coordination I: Delegation & Topologies

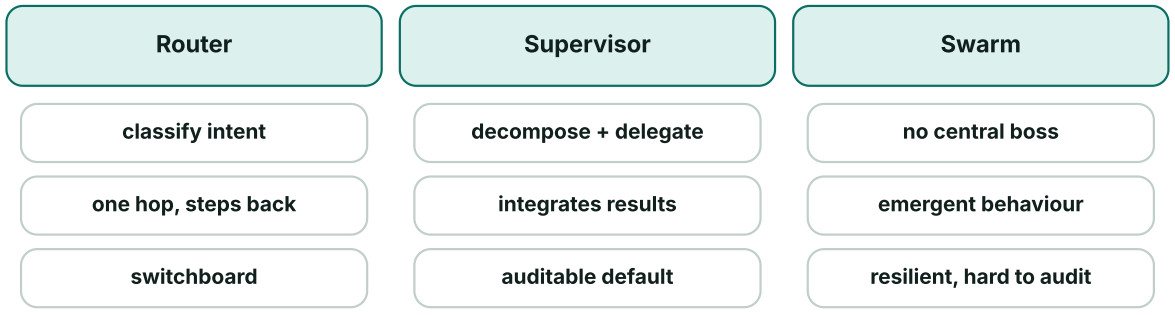
1. The principle

Every coordination pattern answers one question: **who does what, and who decides?** Use the **least coordination** that solves the problem, because every mechanism adds latency, cost and failure surface. Start structured; earn decentralisation.

2. Router, supervisor, swarm

- **Agent Router**: a lightweight classifier reads the request's intent and dispatches it to the right specialist, then steps back. Single point of failure; needs a fallback for ambiguous requests
- **Supervisor**: a central orchestrator breaks the goal into sub-tasks, delegates, and integrates the results. Predictable and auditable, but a bottleneck and single point of failure. **The enterprise default**
- **Swarm**: many agents follow simple local rules with no boss; global behaviour emerges. Resilient and adaptive, but hard to predict, govern and audit

✎ DIAGRAM



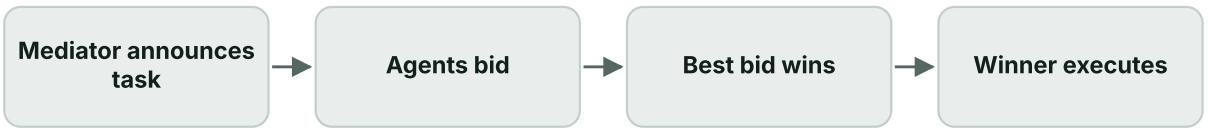
A router picks an agent; a supervisor manages several; a swarm has no manager.

3. Composition topologies

- **Blackboard Knowledge Hub**: agents read and write a shared workspace; a controller invokes whichever specialist can contribute next (e.g. collaborative medical diagnosis). Flexible; shared state can become a contention problem
- **Contract-Net Marketplace**: a mediator announces a task, agents **bid**, the best bid wins (e.g. choosing a cloud provider on cost and latency). Great for dynamic allocation; bidding adds latency
- **Supervision Tree with Guarded Capabilities**: supervisors monitor children and restart or contain failures; each node gets only the capabilities it needs (e.g. a

resilient web scraper). Strong fault isolation and least privilege

✎ DIAGRAM



Contract-Net: dynamic allocation by bidding.

SAY IT

"A router picks an agent and steps back; a supervisor stays in charge and integrates the results. Supervisor is the enterprise default because auditability matters more than emergent flexibility."